

ROLL NO. ...

Rawal Institute of Engineering & Technology, Faridabad
Sessional Exam I (February, 2024)

Semester: II

Branch: CSE

Subject: Mathematics

Paper Code: BSC106E

Time: 90Mins

Max. Marks: 30

Note: Part A is compulsory. Attempt any three questions from Part B.

PART-A

I.

- Find the probability of drawing a card from a well shuffled pack such that the drawn card is either a king or queen.
- Find the second moment of Binomial distribution about mean.
- Find the m.g.f of a random variable whose moments are $\mu_r' = (r+1)! \cdot 2^r$
- What is probability density function for Normal distribution?
- If $f(x) = \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}x^2}$, then its mean is
- Show _____.

that $f(x) = \begin{cases} \frac{\alpha^\lambda e^{-\alpha x} x^{\lambda-1}}{\Gamma(\lambda)}, & x \geq 0, \alpha, \lambda > 0 \\ 0, & \text{otherwise} \end{cases}$ is a p.d.f.

PART-B

- Show that Gamma Distribution tends to normal distribution for large values of parameter λ .
- Suppose 10% of probability for a normal distribution $N(\mu, \sigma^2)$ is below 25 and 5% above 90, what are the value of μ and σ .
- If $f(x) = ke^{-|x|}$ is p.d.f in $-\infty < x < \infty$ find the values of k and variance of the random variable and also find the probability between 0 and 4.
- Let X have the p.d.f: $f(x) = \frac{1}{2\sqrt{3}}, -\sqrt{3} < x < \sqrt{3}$ and $f(x) = 0$, elsewhere. Find the actual probability $P[|X - \mu| \leq \frac{3}{2}\sigma]$ and compare it with the upper bound obtained by Chebychev's inequality.

Part - A

- 1 (i) Let A = Event of the king card drawn
 B = Event of the queen card drawn

$$P(A) = \frac{4}{52}, \quad P(B) = \frac{4}{52}$$

Probability that the drawn card is either a king or queen is.

$$P(A+B) = P(A) + P(B) = \frac{4}{52} + \frac{4}{52} = \frac{8}{52} = \frac{2}{13}$$

- (ii) The Binomial distribution is given by
 $P(X=x) = {}^nC_x p^x q^{n-x}$, $x = 0, 1, 2, \dots, n$ and $p+q=1$

$$K'_2 = \sum_{x=0}^n x^2 P(X=x)$$

$$= \sum_{x=0}^n (x(x-1) + x) {}^nC_x p^x q^{n-x}$$

$$= \sum_{x=0}^n x(x-1) {}^nC_x p^x q^{n-x} + \sum_{x=0}^n x \cdot {}^nC_x p^x q^{n-x}$$

$$= [2 \cdot 1 \cdot {}^nC_2 p^2 q^{n-2} + 3 \cdot 2 \cdot {}^nC_3 p^3 q^{n-3} + \dots + p^{n-2}] + E(X)$$

$$= n(n-1) p^2 [q^{n-2} + (n-2)pq^{n-3} + \dots + p^{n-2}] + np$$

$$= n(n-1) p^2 (q+p)^{n-2} + np$$

$$K'_2 = n(n-1) p^2 + np$$

$$K_2 = K'_2 - (K'_1)^2 = n(n-1)p^2 + np - n^2 p^2$$

$$= npq$$

- (iii) The m.g.f is given by

$$M_X(t) = \sum_{x=0}^{\infty} \frac{t^x}{x!} K'_x = \sum_{x=0}^{\infty} \frac{t^x}{x!} (x+1)! p^x$$

$$= \sum_{x=0}^{\infty} (x+1) (2t)^x$$

$$M_X(t) = 1 + 2 \cdot (2t) + 3(2t)^2 + 4(2t)^3 + \dots$$

$$= (1-2t)^{-2}$$

(iv) The continuous random variable X is said to have a normal distribution, if its probability density function is defined as

$$f(x) = \frac{1}{\sigma \sqrt{2\pi}} e^{-\frac{1}{2} \left(\frac{x-\mu}{\sigma} \right)^2}, \quad -\infty \leq x \leq \infty, \sigma > 0$$

where μ and σ are parameters of the normal distribution.

(v) when $f(x) = \frac{1}{\sigma \sqrt{2\pi}} e^{-\frac{1}{2} \left(\frac{x-\mu}{\sigma} \right)^2}$ then mean is μ

put $\mu = 0$ and $\sigma = 1$

$$f(x) = \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2} x^2}$$

$$\therefore \boxed{\mu = 0}$$

$$\begin{aligned} \text{Mean} = E(X) &= \int_{-\infty}^{\infty} x f(x) dx \\ E(X) &= \int_{-\infty}^{\infty} x \cdot \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2} x^2} dx \\ E(X) &= 0 \end{aligned}$$

$\left\{ \begin{array}{l} x e^{-\frac{1}{2} x^2} \\ \text{is an} \\ \text{odd function} \end{array} \right.$

$$(vi) \int_{-\infty}^{\infty} f(x) dx = \int_{-\infty}^0 f(x) dx + \int_0^{\infty} f(x) dx.$$

$$= \int_0^{\infty} f(x) dx \quad \{ \because x \geq 0 \}$$

$$= \int_0^{\infty} \frac{\kappa^{\lambda} e^{-\kappa x} x^{\lambda-1}}{\Gamma(\lambda)} dx$$

$$= \frac{\kappa^{\lambda}}{\Gamma(\lambda)} \int_0^{\infty} e^{-\kappa x} x^{\lambda-1} dx$$

$$= \frac{\kappa^{\lambda}}{\Gamma(\lambda)} \cdot \frac{\Gamma(\lambda)}{\kappa^{\lambda}} = 1 \quad (\text{using gamma function}).$$

II. Limiting form of Gamma Distribution as $\lambda \rightarrow \infty$

If X follows gamma distribution, then $E(X) = \frac{\lambda}{\kappa}$ and $\text{Var}(X) = \frac{\lambda}{\kappa^2}$. Then standard gamma variate is $Z = \frac{X - \lambda/\kappa}{\sqrt{\lambda}/\kappa}$

$$\therefore Z = \frac{X - \lambda/\kappa}{\frac{\sqrt{\lambda}}{\kappa}} = \frac{X}{\left(\frac{\sqrt{\lambda}}{\kappa}\right)} - \sqrt{\lambda}$$

$$\begin{aligned} M_Z(t) &= E[e^{Zt}] \\ &= E\left[e^{t\left(\frac{X}{\left(\frac{\sqrt{\lambda}}{\kappa}\right)} - \sqrt{\lambda}\right)}\right] \\ &= e^{-t \cdot \sqrt{\lambda}} E\left[e^{t(X/(\sqrt{\lambda}/\kappa))}\right] \\ &= e^{-t \cdot \sqrt{\lambda}} M_X(t/(\sqrt{\lambda}/\kappa)) \end{aligned}$$

$$M_Z(t) = e^{-t \cdot \sqrt{\lambda}} \left(1 - \frac{t}{\sqrt{\lambda}}\right)^{-\lambda}, \quad t < \sqrt{\lambda}$$

$$\begin{aligned} \text{Also } K_Z(t) &= -\sqrt{\lambda} \cdot t - \lambda \log\left(1 - \frac{t}{\sqrt{\lambda}}\right) \\ &= -\sqrt{\lambda} \cdot t + \lambda \left(\frac{t}{\sqrt{\lambda}} + \frac{t^2}{2\lambda} + \frac{t^3}{3\lambda^{3/2}} + \dots\right) \end{aligned}$$

$$= -\sqrt{\lambda} \cdot t + \sqrt{\lambda} \cdot t + \frac{t^2}{2} + O(\lambda^{-1/2})$$

where $O(\lambda^{-1/2})$ are terms containing $\frac{1}{2}$ and higher powers of λ in the denominator

$$\lim_{\lambda \rightarrow \infty} K_Z(t) = \frac{t^2}{2} \Rightarrow \lim_{\lambda \rightarrow \infty} M_Z(t) = e^{t^2/2}$$

which is the m.g.f of a standard normal variate.
 $\therefore$ gamma distribution tends to normal distribution for large values of parameter λ .

III.

$$P(X \leq 35) = \frac{10}{100} = 0.1$$

(4)

$$P(X > 90) = \frac{5}{100} = 0.05$$

Standard normal variate $Z = \frac{X - \mu}{\sigma}$

$$\text{When } X = 35 \text{ then } Z = \frac{35 - \mu}{\sigma}$$

$$\text{When } X = 90 \text{ then } Z = \frac{90 - \mu}{\sigma}$$

Now $P(X \leq 35) = 0.1$

$$P(Z \leq \frac{35 - \mu}{\sigma}) = 0.1$$

$$\frac{35 - \mu}{\sigma} = z \quad (\because \text{value is not given})$$

$$\mu + z \cdot \sigma = 35 \quad \text{--- (1)}$$

Now $P(X > 90) = 0.05$

$$P(Z > \frac{90 - \mu}{\sigma}) = 0.05$$

$$\frac{90 - \mu}{\sigma} = \beta$$

$$\mu + \beta \cdot \sigma = 90 \quad \text{--- (2)} \quad \left\{ \because \text{value is not given} \right\}$$

for suitable value of z & β (if given or from table)
we can find μ & σ .

(5)

$$\int_{-\infty}^{\infty} f(x) dx = 1 \Rightarrow \int_{-\infty}^{\infty} K e^{-|x|} dx = 1$$

$$\int_{-\infty}^0 K e^{-|x|} dx + \int_0^{\infty} K e^{-|x|} dx = 1$$

$$\int_{-\infty}^0 K e^x dx + \int_0^{\infty} K e^{-x} dx = 1$$

$$\left\{ \because |x| = \begin{cases} -x & \text{if } x < 0 \\ x & \text{if } x > 0 \end{cases} \right\}$$

$$\therefore K [e^x]_{-\infty}^0 + K [e^{-x}]_0^{\infty} = 1$$

$$\Rightarrow K [1 - 0] - K (0 - 1) = 1$$

$$2K = 1 \Rightarrow K = \frac{1}{2}$$

$$\therefore f(x) = \frac{1}{2} e^{-|x|}$$

$$E(x) = \int_{-\infty}^{\infty} x f(x) dx$$

$$= \int_{-\infty}^{\infty} x \cdot \frac{1}{2} e^{-|x|} dx = 0 \quad \left\{ \because x e^{-|x|} \text{ is an odd function} \right\}$$

$$E(x^2) = \int_{-\infty}^{\infty} x^2 f(x) dx = \int_{-\infty}^{\infty} x^2 \cdot \frac{1}{2} e^{-|x|} dx$$

$$= \frac{1}{2} \int_{-\infty}^{\infty} x^2 e^{-|x|} dx = \frac{1}{2} \cdot 2 \int_0^{\infty} x^2 e^{-x} dx$$

$$= \int_0^{\infty} x^2 e^{-x} dx$$

$$= \left[x^2 \frac{e^{-x}}{-1} - 2x \frac{e^{-x}}{-1} + 2 \frac{e^{-x}}{-1} \right]_0^{\infty} = 2$$

$$\text{Var}(x) = E(x^2) - (E(x))^2 = 2$$

$$\text{Now } P(0 \leq x \leq 4) = \int_0^4 f(x) dx = \int_0^4 \frac{1}{2} e^{-x} dx = \frac{1}{2} \int_0^4 e^{-x} dx$$

$$= \frac{1}{2} \left[\frac{e^{-x}}{-1} \right]_0^4$$

$$= \frac{1}{2} (e^{-4} - e^0)$$

$$= \frac{1}{2} (1 - e^{-4}) = 0.491$$

(v).

$$E(X) = \int_{-\sqrt{3}}^{\sqrt{3}} \frac{x}{2\sqrt{3}} dx = 0 \quad \left\{ \because x \text{ is an odd function} \right\}$$

(6)

$$E(X^2) = \int_{-\sqrt{3}}^{\sqrt{3}} \frac{x^2}{2\sqrt{3}} dx = \frac{1}{6\sqrt{3}} (3\sqrt{3} + 3\sqrt{3}) = 1$$

$$\sigma^2 = \text{Var}[X] = E(X^2) - (E(X))^2 = 1$$

By Chebyshev's Inequality

$$P\left[|X - \mu| \geq \frac{3}{2}\sigma\right] \leq \frac{\sigma^2}{\frac{9}{4}} = \frac{4}{9}$$

Actual Probability

$$\begin{aligned} P\left[|X - \mu| \geq \frac{3}{2}\sigma\right] &= P\left[|X - 0| \geq \frac{3}{2}\sigma\right] \\ &= P\left[|X| \geq \frac{3}{2}\sigma\right] \\ &= 1 - P\left[|X| < \frac{3}{2}\right] \\ &= 1 - P\left[-\frac{3}{2} < X < \frac{3}{2}\right] \\ &= 1 - \int_{-3/2}^{3/2} \frac{dx}{2\sqrt{3}} \\ &= 1 - \frac{1}{2\sqrt{3}} \left(\frac{3}{2} + \frac{3}{2}\right) \\ &= 1 - \frac{3}{2\sqrt{3}} = \frac{2 - \sqrt{3}}{2} \end{aligned}$$

Hence, the actual probability is $\frac{2 - \sqrt{3}}{2}$
 whereas the upper bound obtained by Chebyshev's Inequality is $\frac{4}{9}$